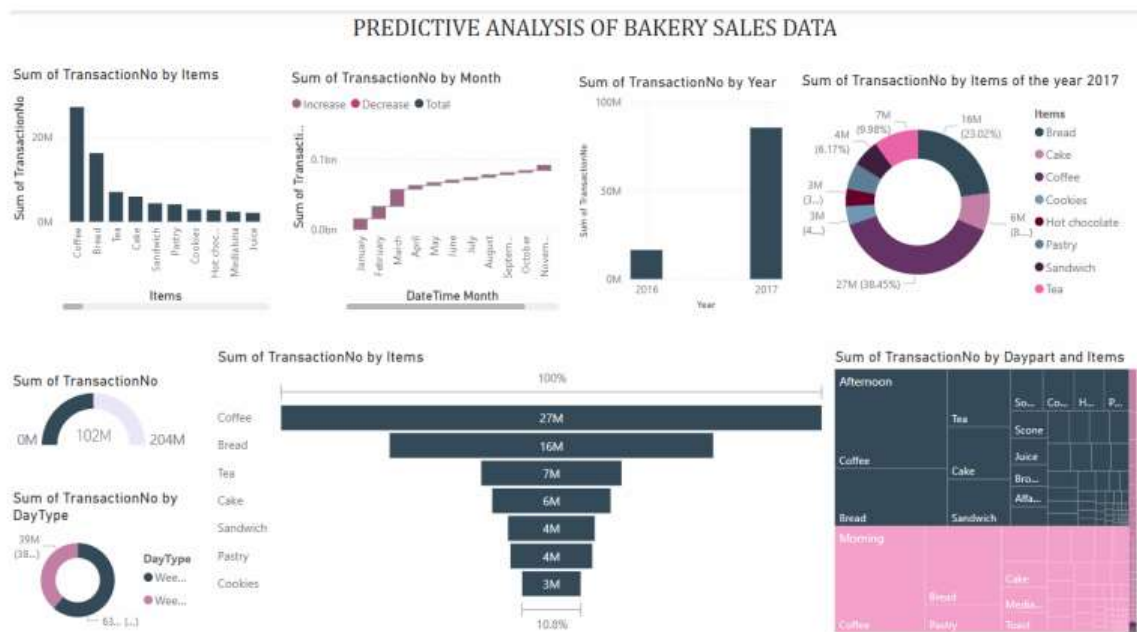


PREDICTIVE ANALYSIS OF BAKERY SALES DATA

Overview

This analysis examines bakery sales transactions across multiple dimensions, including product category, month, year, day type, and time of day, to identify customer purchasing patterns and provide actionable business insights. The dataset covers the period from 2016 to 2017 and represents a total transaction volume exceeding 204 million units. The findings highlight key demand trends, operational opportunities, and potential strategies to improve sales performance and customer retention.

Analysis Dashboard



Key Findings

- **Morning and evening periods generate the highest sales volumes**, making them the bakery's most important revenue-generating time slots.
- **Transaction activity consistently declines during the mid-day period (12 PM–3 PM)**, suggesting that maintaining full staffing levels and inventory during these hours may lead to operational inefficiencies.
- **Sales performance improved significantly in 2017 compared to 2016**, with Hot Chocolate emerging as the best-performing product, indicating a shift in customer preferences and seasonal demand patterns.

- **Weekday sales substantially exceed weekend sales** (approximately 63 million versus 39 million transactions), highlighting a strong commuter-driven customer base and predictable purchasing behavior.
- **Coffee, Bread, and Tea remain the bakery's core products**, collectively accounting for more than 70% of total sales volume and serving as the primary drivers of revenue.

Strategic Recommendations

1. Time-Based Menu Optimization

Introduce targeted menu offerings during peak morning and evening hours, such as seasonal beverages, flavored coffee options, fresh snacks, and specialty items. Aligning product availability with customer demand can enhance sales while reducing food waste and improving operational efficiency.

2. Operational Resource Optimization

Consider adjusting staffing schedules during the lower-demand period between 12 PM and 3 PM. Staff can be reassigned to preparation, inventory management, restocking, and quality control activities. Employing part-time staff during peak periods may also help optimize labor costs while maintaining service quality.

3. Seasonal Product Rotation

Implement a structured product rotation strategy by introducing new items every two months based on seasonal preferences. Examples include cold beverages and lighter menu options during summer months, and spiced drinks or comfort foods during colder seasons. This approach can sustain customer interest and encourage repeat visits.

4. Strengthen Customer Loyalty Programs

Given the strong weekday customer base, introducing a loyalty card or app-based rewards program could increase customer retention and repeat purchases. Targeting regular morning customers with personalized offers and incentives may drive long-term revenue growth. Even a modest increase in repeat purchase rates can have a meaningful impact on overall sales performance.